Challenger O-Ring Data – Logistic Regression

Code

Background

Watch the Introductory Movie (http://www.history.com/topics/challenger-disaster/videos/engineering-disasters---challenger)

Code

	Temp	Pres	Fail	n	Erosion	BlowBy	Damage	Date
1	66	50	0	6	0	0	0	4/12/81
2	70	50	1	6	1	0	4	11/12/81
3	69	50	0	6	0	0	0	3/22/82
5	68	50	0	6	0	0	0	11/11/82
6	67	50	0	6	0	0	0	4/4/83
7	72	50	0	6	0	0	0	6/18/83

As described in the video, it was known that the o-rings could fail when the tempurature was below 53° F. It then stated that “Nasa had already launched shuttles below 53° and gotten away with it.” This second statement is actually false. The lowest temperature of any of the 23 prior launches (before the Challenger explosion) was 53° F [view source] (http://www.ics.uci.edu/~staceyah/111-202/handouts/Dalal_etal_1989-Challenger.pdf). The “evidence” that the o-rings could fail below 53° was based on a simple conclusion that since the launch at 53° experienced two o-ring failures, it seemed unwise to launch below that temperature. The statement that NASA had launched previously with o-ring failures occurring, but the launch still being successful was true, but all launches had a temperature of at least 53° F. Thus, whether or not temperature was the root cause of the failures was debatable.

Logistic Model

To model the probability of the o-rings failing at various temperatures, we could apply the logistic regression model

$$P(Y_i = 1|x_i) = \frac{e^{\beta_0+\beta_1 x_i}}{1+e^{\beta_0+\beta_1 x_i}} = \pi_i$$

where for observation i:

- $Y_i = 1$ denotes at least one o-ring failing for a given launch,
- $Y_i = 0$ denotes no o-rings failing (successful launch), and
- x_i denotes the outside temperature in degrees farenheit at the time of the launch.

Note that if β_1 is zero in the above model, then x_i (temperature) provides no insight about the probability of a failed O-ring. Thus, we could test the hypothesis that

$$H_0 : \beta_1 = 0$$
$$H_a : \beta_1 \neq 0$$

Fitting the Model

To obtain estimates of the coefficients β_0 and β_1 for the `challeng` data, we apply the following R code.

Code

```
##
## Call:
## glm(formula = Fail > 0 ~ Temp, family = binomial, data = challeng)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -1.0611  -0.7613  -0.3783   0.4524   2.2175
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  15.0429     7.3786   2.039   0.0415 *
## Temp         -0.2322     0.1082  -2.145   0.0320 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 28.267  on 22  degrees of freedom
## Residual deviance: 20.315  on 21  degrees of freedom
## AIC: 24.315
##
## Number of Fisher Scoring iterations: 5
```

Thus the estimated model for π_i is given by

$$P(Y_i = 1|x_i) \approx \frac{e^{15.043-0.232x_i}}{1+e^{15.043-0.232x_i}} = \hat{\pi}_i$$

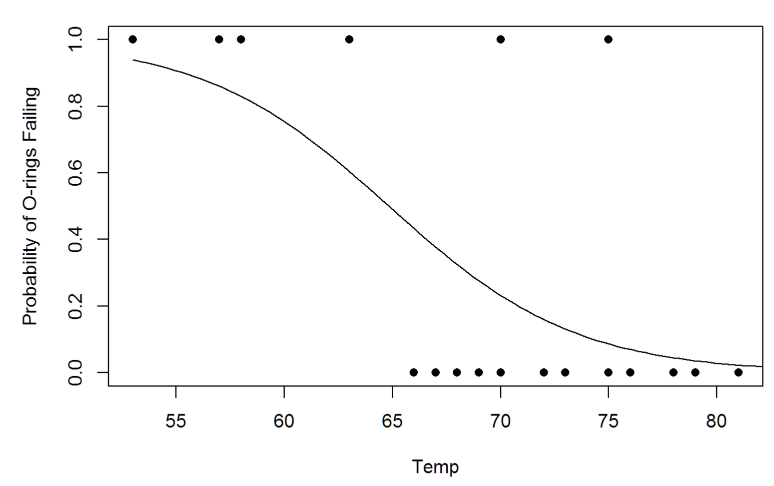
where $b_0 = 15.043$ is the value of the `(Intercept)` which estimates β_0 and $b_1 = -0.232$ is the value of `Temp` which estimates β_1.

Importantly, the p-value for the test of `Temp` shows a significant result ($p = 0.0320$) giving sufficient evidence to conclude that $\beta_1 \neq 0$, which allows us to conclude that temperature effects the probability of an O-ring failure.

Visualizing the Model

To visualize this simple logistic regression we could make the following plot.

Code



Diagnosing the Model

To demonstrate that the logistic regression is a good fit to these data we apply the Hosmer-Lemeshow goodness of fit test (since there are only a couple repeated x-values) from the `library(ResourceSelection)`.

Code

Code

```
##
##  Hosmer and Lemeshow goodness of fit (GOF) test
##
## data:  chall.glm$y, chall.glm$fitted
## X-squared = 7.4118, df = 4, p-value = 0.1157
```

Code

Recall that the null hypothesis is that the logistic regression is a good fit for the data, thus we conclude the null, and claim that the logistic regression is appropriate.

Interpreting the Model

Since the temperature being zero is not really realistic for this model, the value of e^{b_0} is not interpretable. However, the value of $e^{b_1} = e^{-0.232} \approx 0.79$ shows that the odds of the o-rings failing for a given launch decreases and changes by a factor of 0.79 for every 1° F increase in temperature. (Note that this implies that every 1° F decrease in temperature increases the odds of a failed o-ring by a factor of $e^{0.232} \approx 1.26$.) The Challenger shuttle was launched at a tempurature of 31° F. By waiting until 53° F, the odds of failure would have been decreased by a factor of $e^{-0.232(53-31)} \approx 0.006$. Note that for a temperature of 31° F our model puts the probability of a failure at

$$P(Y_i = 1|x_i) \approx \frac{e^{15.043-0.232\cdot 31}}{1+e^{15.043-0.232\cdot 31}} = \hat{\pi}_i$$

which, using R to do this calculation we get $\hat{\pi}_i \approx$

Code

```
##         1
## 0.9996088
```